

Lab Exercise 04: Introduction to Wireshark

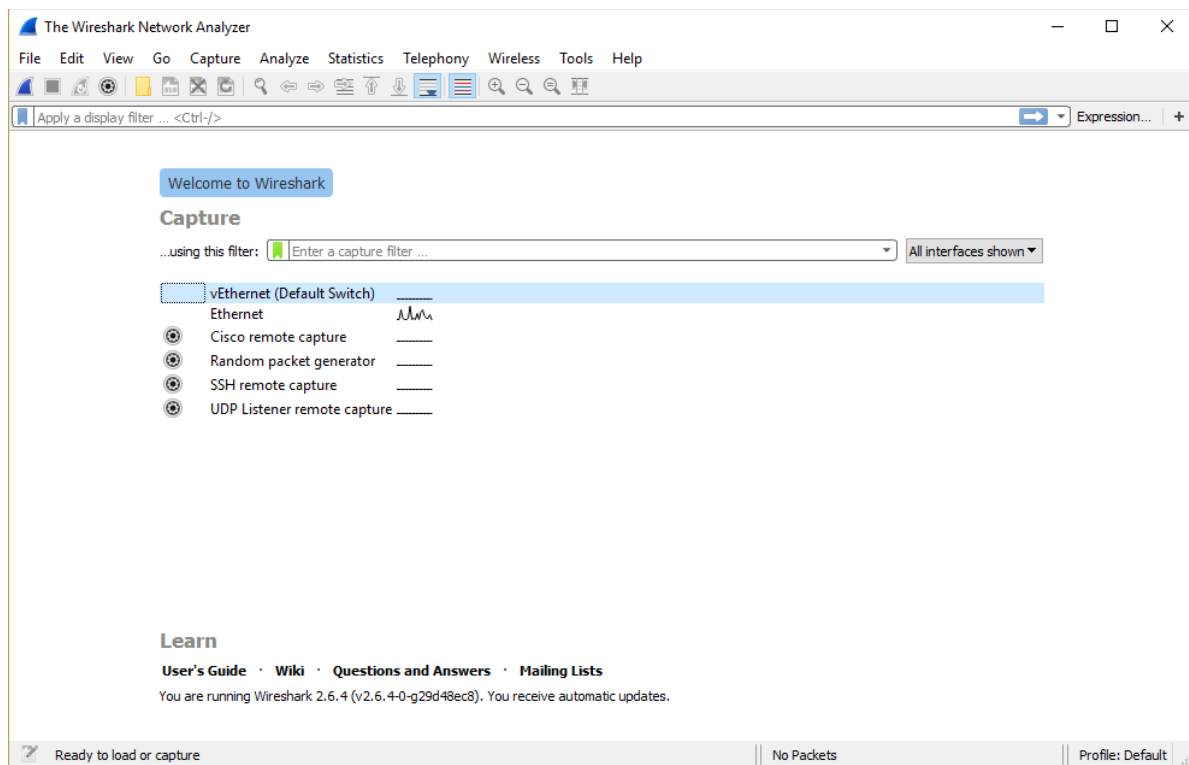
BACKGROUND:

Wireshark is a popular free network security analysis tool that can be used to inspect traffic, protocol data, identify denial-of-service attacks and troubleshoot devices and firewalls.

In this tutorial, you will learn how to use this tool to capture network traffic.

1. RUNNING WIRESHARK:

Go to **Apps Anywhere** application and after validation process locate Wireshark using the search facility and launch it. A welcome screen similar to the one shown below should be visible

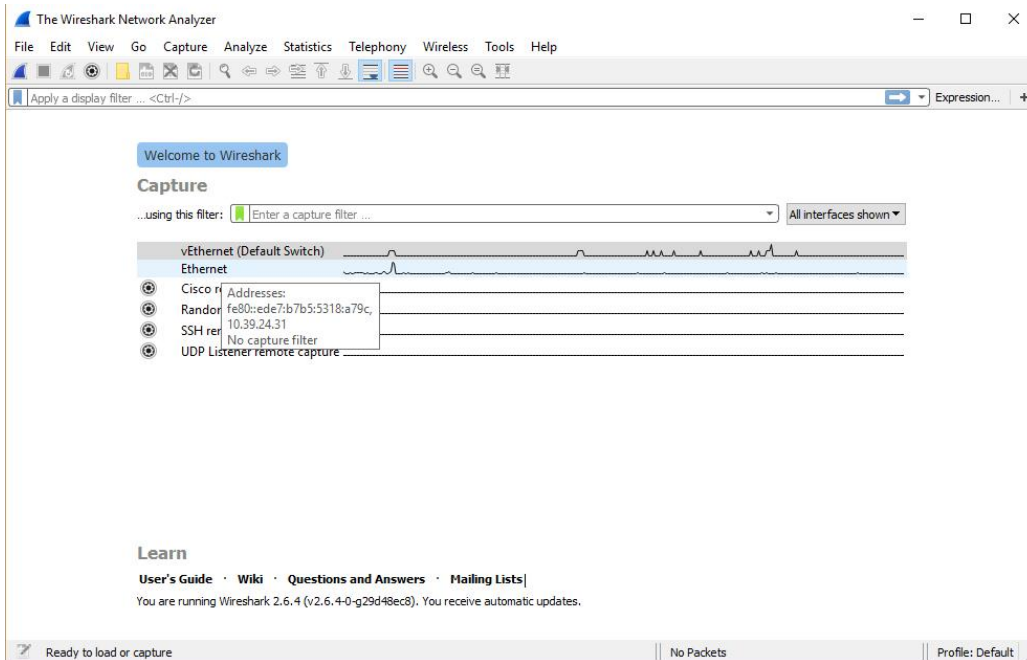


This screen contains a list of available network connections on your PC. In this example above, the following connection types are shown: *vEthernet (virtual Ethernet)*, *Ethernet*, *Cisco Remote Capture*, *Random Packet Generator*, *SSH remote capture*, *UDP Listener Remote Capture*. Live traffic on a network connection is depicted as an ECG-style line graph displayed to the right of each respective network.

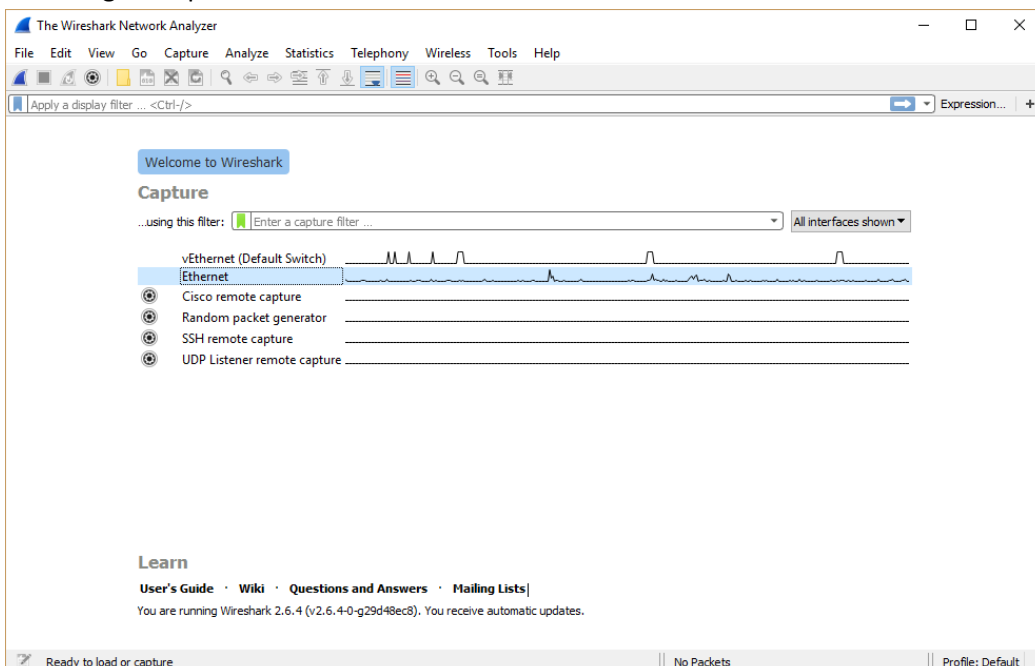
What network connections do you have on your PC? Which of them has live traffic?

2. Capturing Packets

First, move the mouse over the network connection and see how simple information is provided. In the example below when the mouse hovers on the Ethernet, the IPv6 and IPv4 addresses and information about the capture filter.

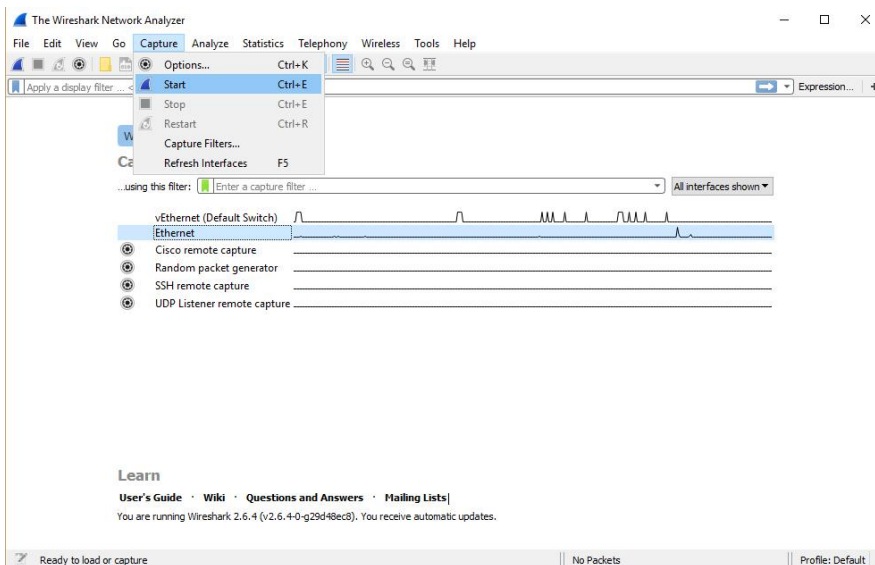


Then select one of these networks providing live traffic by clicking on it. Once a connection type is selected for capturing purposes, its background will be shaded in either blue or gray. Ethernet connection is selected in the following example:



Note that if you'd like to record data from multiple networks simultaneously you need to use the Shift or Ctrl keys when you are clicking.

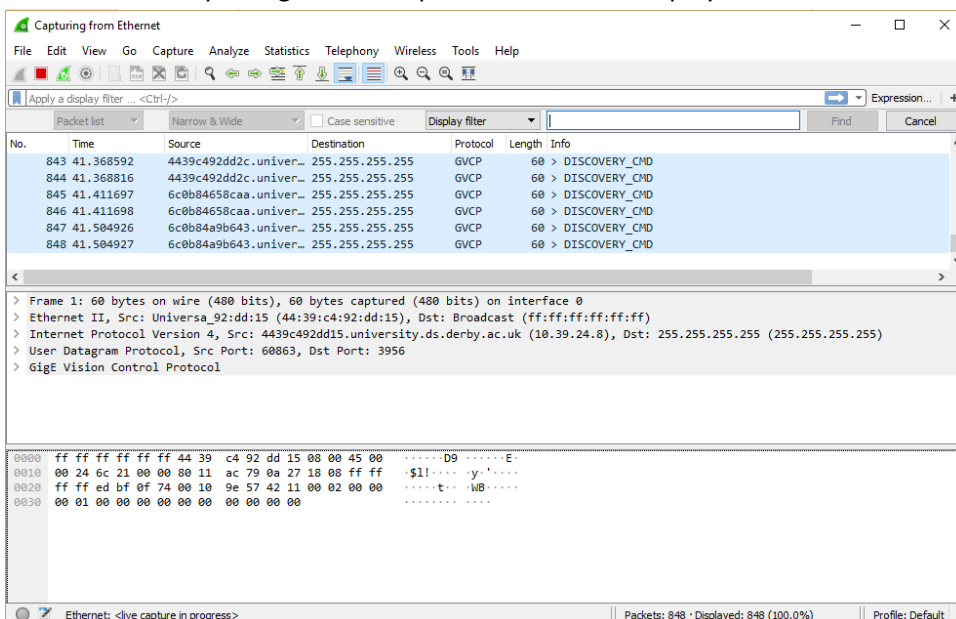
Now click on **Capture** from the main menu, located towards the top of the Wireshark interface. When the drop-down menu appears, select the **Start** option.



There are other shortcuts to start packet capturing:

- **Keyboard:** Press Ctrl + E
- **Mouse:** To begin capturing packets from one particular network, simply double-click on its name
- **Toolbar:** Click on the blue shark fin button, located on the far left-hand side of the Wireshark toolbar

When the live capturing starts the packet details are displayed in the Wireshark window as they are recorded:

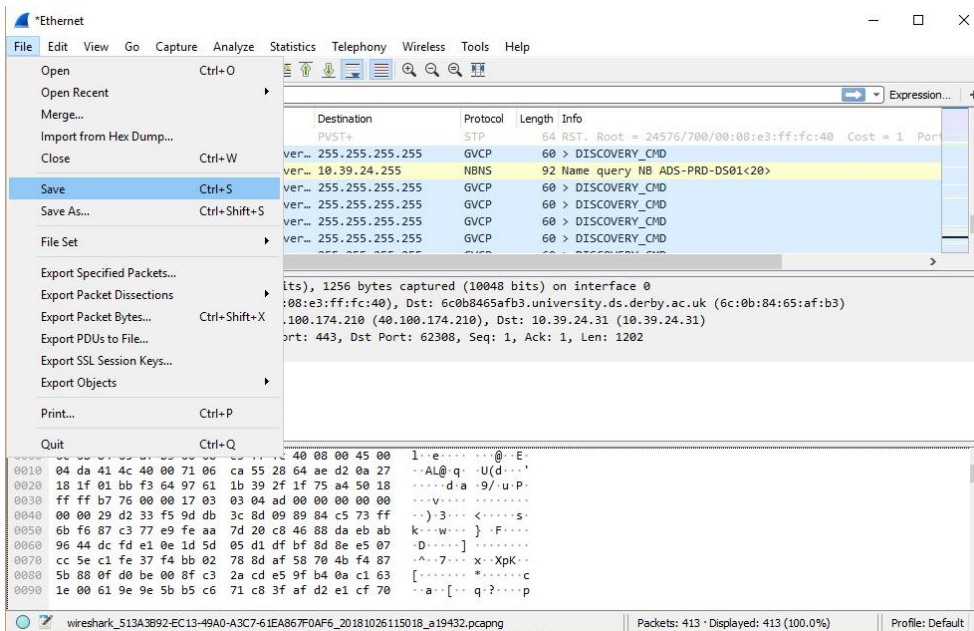


Now the traffic and related data are displayed on the main screen.

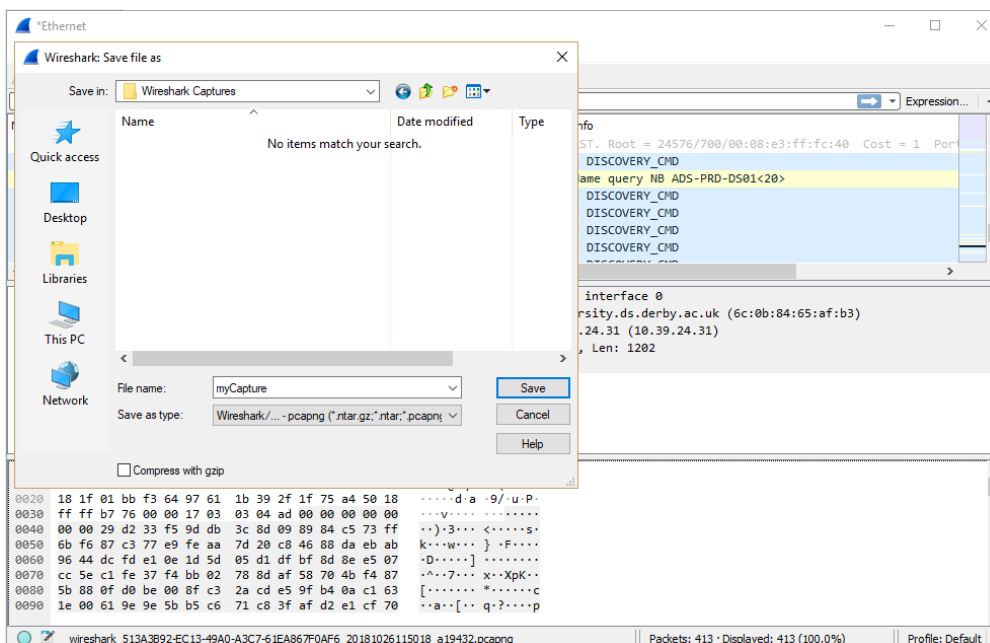
To stop capturing perform one of the actions below:

- **Keyboard:** Press **Ctrl + E**
- **Toolbar:** Click on the red stop button, located next to the shark fin on the Wireshark toolbar

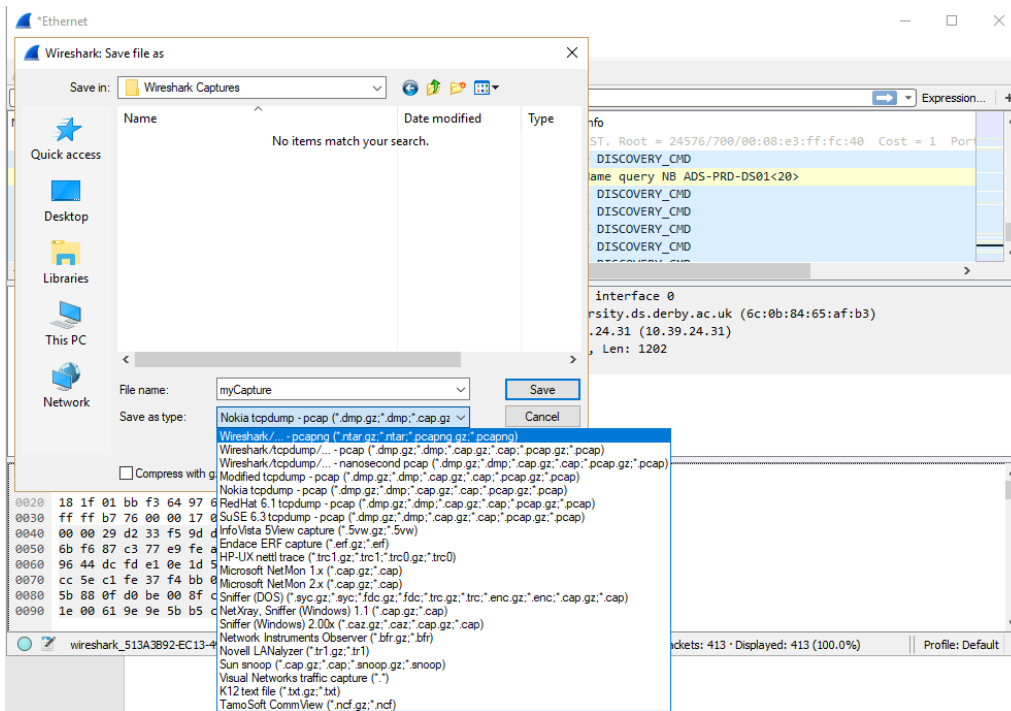
You can save this session using the **Save** option in the **File** menu (Keyboard shortcut is **Ctrl+S**):



In the Save dialog box, navigate to an appropriate folder, type an appropriate file name and click on the **Save** button:

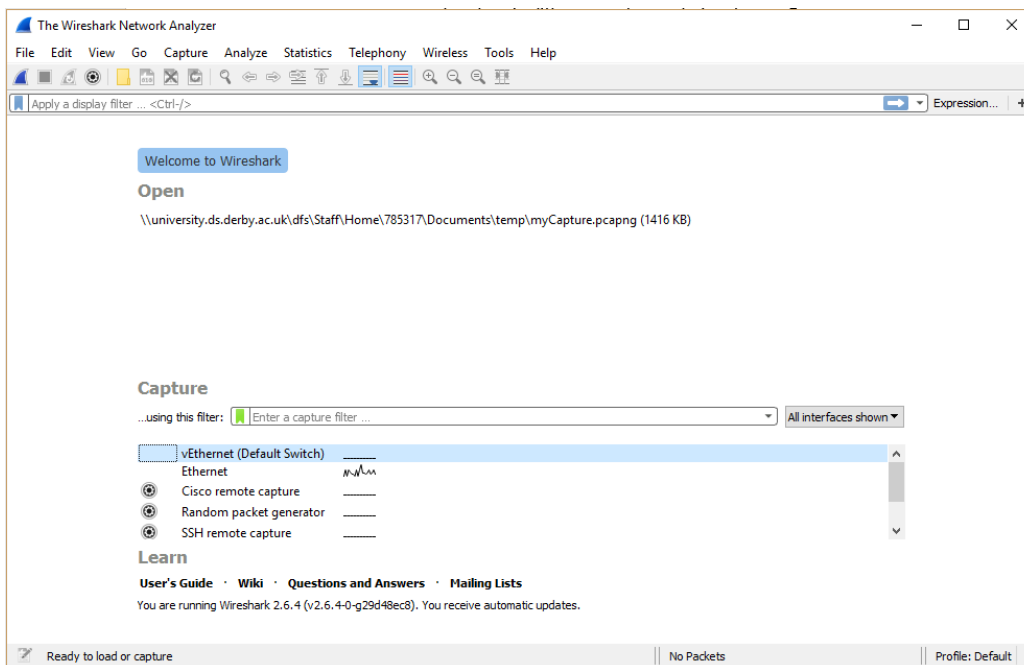


Wireshark Default file type is “pcapng” but you can select other file types by changing the “**Save as type**” if you want to use captured packets in different packet analysis software:



Save your recent capture with different types using File -> Save As and check the differences using Explorer. Can you open them using any text editor like Notepad? Do you spot different and similar data in different types?

You can upload previously captured packets to Wireshark. First, close the Wireshark and restart it again. You will see the recently saved captured packet files in the welcome screen:



If you double-click on the file displayed in the **Open** list it is uploaded to the Wireshark window. Please note that you can use **File -> Open** Menu and **Ctrl + O** keyboard shortcut to open a file.

3. Viewing and Analyzing Packet Contents

When you stop the current packet capturing process or you open a file for already captured packets you can inspect and analyse the captured packets. The captured data interface contains three main sections: The packet list pane, the packet details pane, and the packet bytes pane:

The image shows the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with icons for various functions. A display filter bar is present with the text 'Apply a display filter: <Ctrl-/>'. The main interface is divided into three panes, each labeled with a blue box and an arrow:

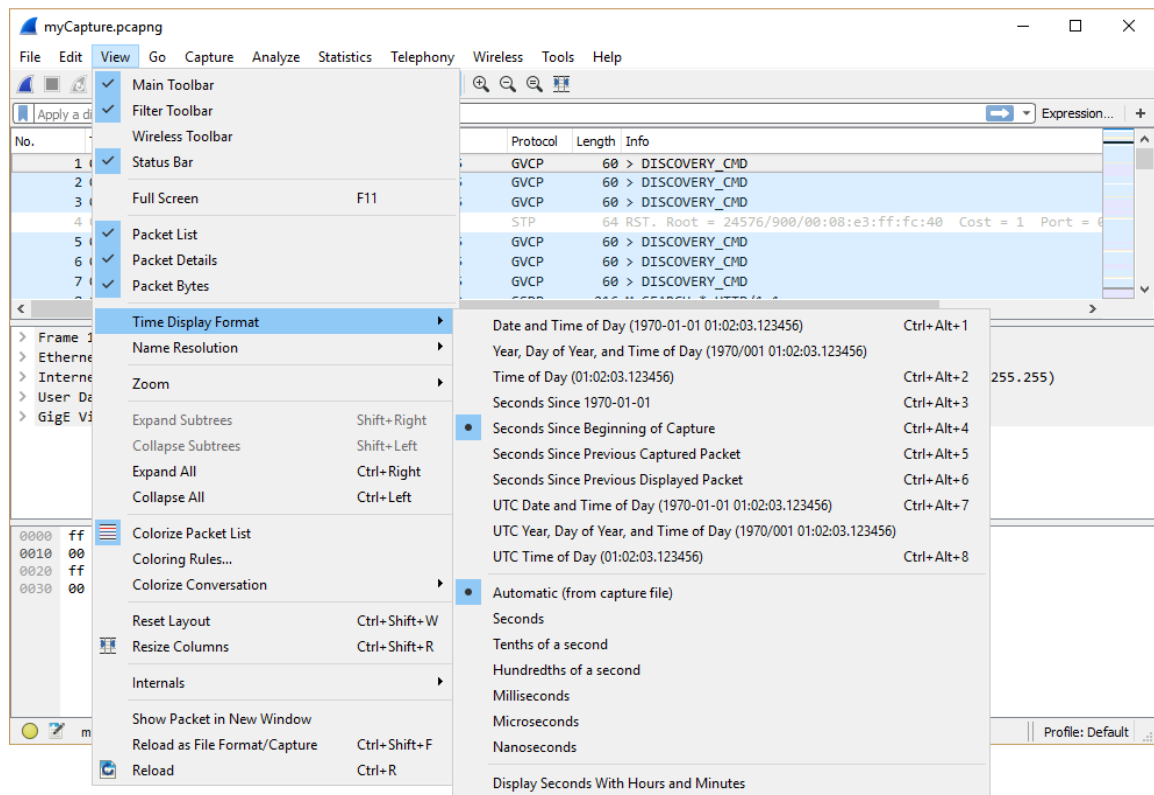
- Packet List Pane:** Displays a list of captured packets. The columns are No., Time, Source, Destination, Protocol, Length, and Info. The first six packets are highlighted in blue.
- Packet Details Pane:** Shows the hierarchical structure of the selected packet (No. 843). It includes details for Ethernet II, Internet Protocol Version 4, User Datagram Protocol, and GigE Vision Control Protocol.
- Packet Bytes Pane:** Displays the raw bytes of the selected packet in hexadecimal and ASCII format.

At the bottom of the interface, a status bar shows 'Ethernet: <live capture in progress>', 'Packets: 848 - Displayed: 848 (100.0%)', and 'Profile: Default'.

Packet List Pane:

The packet list pane shows all packets found in the active capture file. Each row in this pane corresponds to a packet and displays the following data points related to the packet in each column.

- **No:** Number assigned to the packet.
- **Time:** The timestamp of when the packet was captured. This is the relative time since this specific capture file was first created. The default format is the number of seconds (or partial seconds). The time format can be changed using the **Time Display Format** option in the **View Menu**:



Change Time Display format using different options and see the effects in the Packet List Pane.

- **Source:** This is the address (IP or other) where the packet originated.
- **Destination:** This is the address that the packet was sent to.
- **Protocol:** This is the packet's protocol name (TCP, UDP, ARP, etc).
- **Length:** This is the packet length (in bytes).
- **Info:** This shows the details about the packet. The information in this column depends on the packet contents.

When a packet is selected in the top pane, you may notice one or more symbols appear in the first column. Open and/or closed brackets, as well as a straight horizontal line, can indicate whether or not a packet or group of packets are all part of the same back-and-forth conversation on the network. A broken horizontal line signifies that a packet is not part of said conversation.

Click on packets in the top pane and observe grouping.

Double click on a packet. What happens?

Packet Details Pane:

The details pane presents the protocols and protocol fields of the selected packet in a collapsible format. In addition to expanding each selection, you can also apply individual Wireshark filters based on specific details

as well as follow streams of data based on protocol type via the details context menu – accessible by right-clicking your mouse on the desired item within this pane.

Click one of the packets in the Packet List Pane and observe the data in the Packet Details Pane:

The screenshot shows the Wireshark network protocol analyzer interface. The top pane, 'Packet List', displays a list of captured packets. The bottom pane, 'Packet Details', shows the hierarchical structure of the selected packet (Frame 1).

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	4439c492dd15.univer...	255.255.255.255	GVCP	60	> DISCOVERY_CMD
2	0.117420	6c0b84a9b643.univer...	255.255.255.255	GVCP	60	> DISCOVERY_CMD
3	0.117527	6c0b84a9b643.univer...	255.255.255.255	GVCP	60	> DISCOVERY_CMD
4	0.160383	Cisco_15:14e:18		PVST+	64	RST. Root = 24576/900/00:08:e3:ff:fc:40 Cost = 1 Port = 0x8018
5	0.172252	3cd92b4b019d.univer...	255.255.255.255	GVCP	60	> DISCOVERY_CMD
6	0.349199	4439c492dd15.univer...	255.255.255.255	GVCP	60	> DISCOVERY_CMD
7	0.349299	4439c492dd15.univer...	255.255.255.255	GVCP	60	> DISCOVERY_CMD
8	0.467579	e04f4325dc00.univer...	239.255.255.250	SSDP	216	M-SEARCH * HTTP/1.1
9	0.758267	6c0b84a9b643.univer...	Broadcast	ARP	60	Who has 169.254.169.254? Tell 10.39.24.58
10	0.832186	6c0b8465afb3.univer...	bux-fs001.universit...	NBSS	55	NBSS Continuation Message
11	0.838632	bux-fs001.universit...	6c0b8465afb3.univer...	TCP	66	445 → 59097 [ACK] Seq=1 Ack=2 Win=513 Len=0 SLE=1 SRE=2
12	0.894716	3cd92b4b019d.univer...	255.255.255.255	GVCP	60	> DISCOVERY_CMD
13	0.983164	bux-fs001.universit...	6c0b8465afb3.univer...	TCP	60	[TCP Keep-Alive] 445 → 59097 [ACK] Seq=0 Ack=2 Win=513 Len=1

The 'Packet Details' pane for Frame 1 shows the following structure:

- Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0
- Ethernet II, Src: Universa_92:dd:15 (44:39:c4:92:dd:15), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
- Internet Protocol Version 4, Src: 4439c492dd15.university.ds.derby.ac.uk (10.39.24.8), Dst: 255.255.255.255 (255.255.255.255)
- User Datagram Protocol, Src Port: 60863, Dst Port: 3956
- GigE Vision Control Protocol

The packet bytes pane shows the raw data in hexadecimal and ASCII:

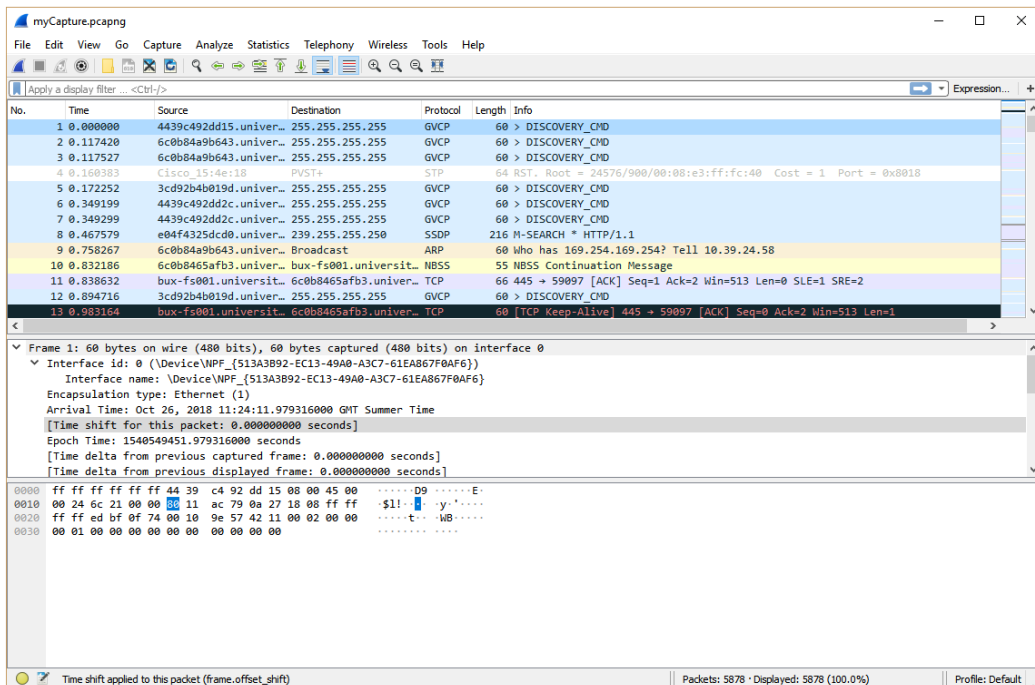
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0000  ff ff ff ff ff ff 44 39 c4 92 dd 15 08 00 45 00  ....D9....E-
0010  00 24 6c 21 00 00 00 11 ac 79 0a 27 18 08 ff ff  ..$!....y'....
0020  ff ff ed bf 0f 74 00 10 9e 57 42 11 00 02 00 00  ....t...WB....
0030  00 01 00 00 00 00 00 00 00 00 00 00 00 00 00  .....

```

Click on the “>” icon at the very left hand side in the Details Pane and check more details:

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Navigate in the details by expanding and collapsing the information.

Packet Bytes Pane:

Packet bytes pane displays the raw data of the selected packet in a hexadecimal view. This hex dump contains 16 hexadecimal bytes and 16 ASCII bytes alongside the data offset.

Selecting a specific portion of this data automatically highlights its corresponding section in the packet details pane and vice versa. Any bytes that cannot be printed are instead represented by a period.

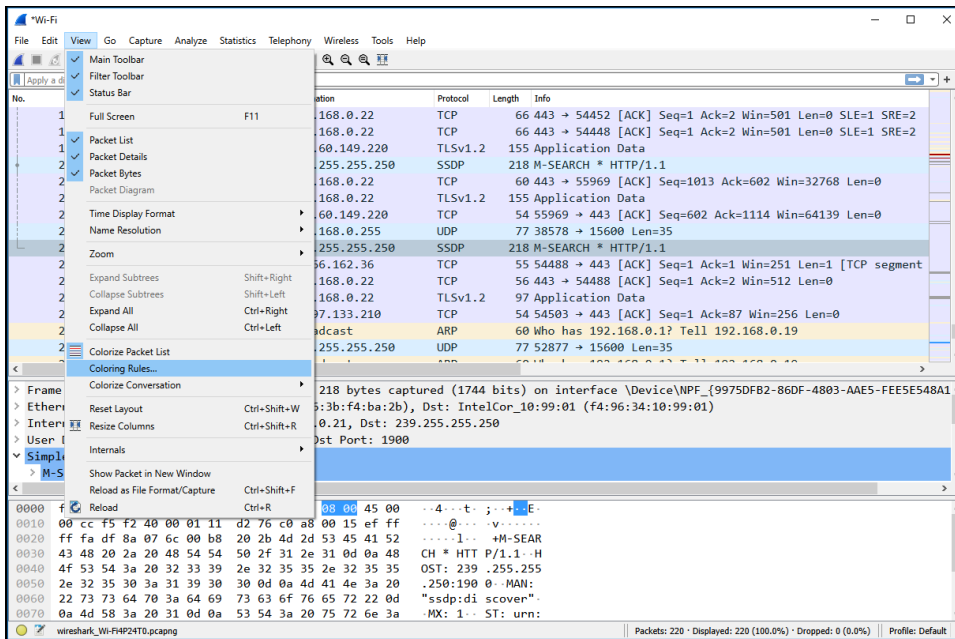
You can choose to show this data in bit format as opposed to hexadecimal by right-clicking anywhere within the pane and selecting the appropriate option from the context menu.

Navigate in the raw details and observe how they have matched the details in the Detail Pane.

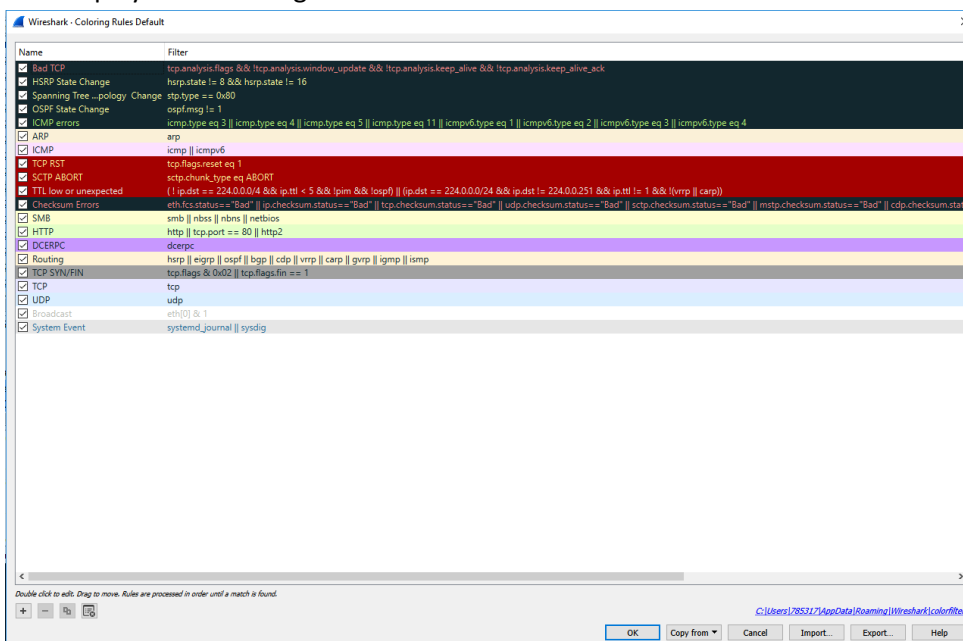
4. Colour Codes

Each row in the Packet List Pane displays a packet captured from the network. Wireshark assigns a different colour to different network traffic. This colour coding is helpful for locating certain packets when you analyse the network traffic. There are 20 default colouring rules built in Wireshark. They can be edited, disabled and deleted. New shade based filters can also be added using Colouring Rules Interface.

You can open the Coloring Rules Interface using **View -> Coloring Rules Menu** :

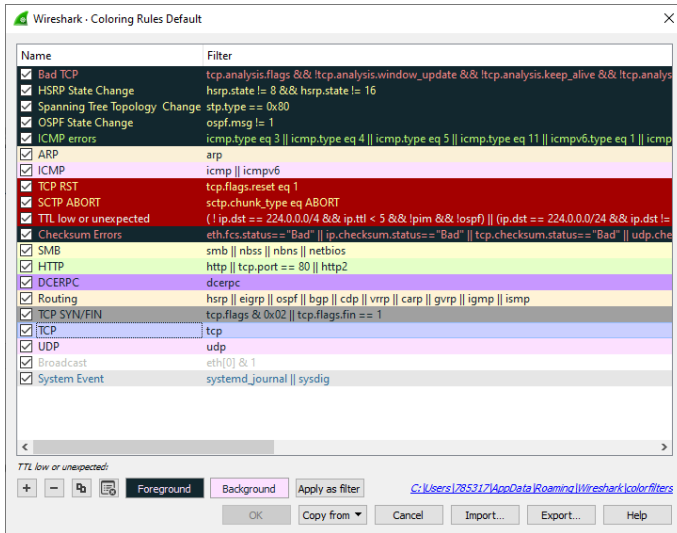


This displays the Coloring Rules Interface:



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You can change the colour of each rule. Click on the TCP rule. Buttons for Foreground and Background will appear at the bottom of the window:

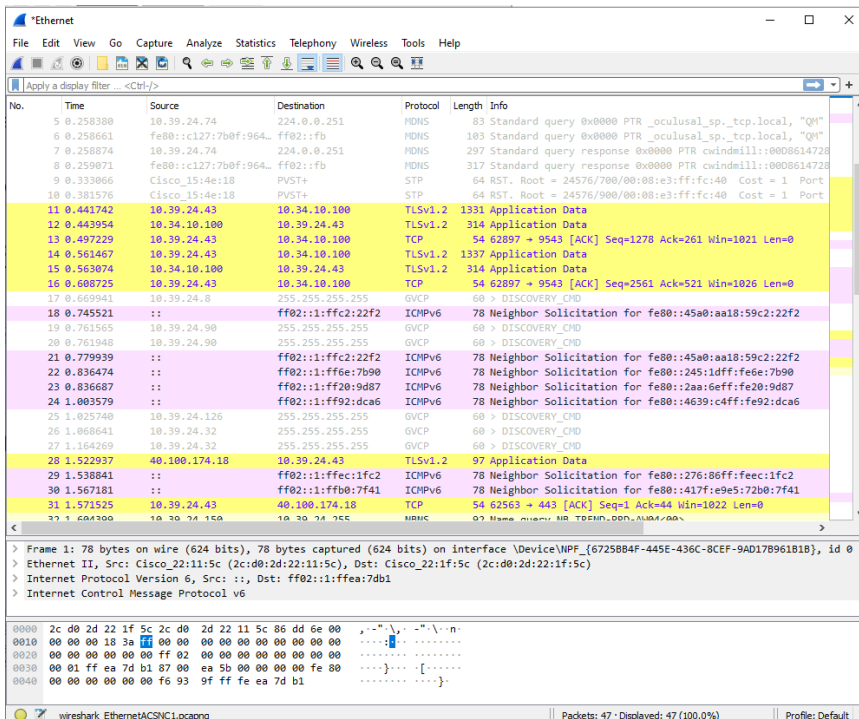


Change Foreground to BLUE and Background to YELLOW. Click on OK.

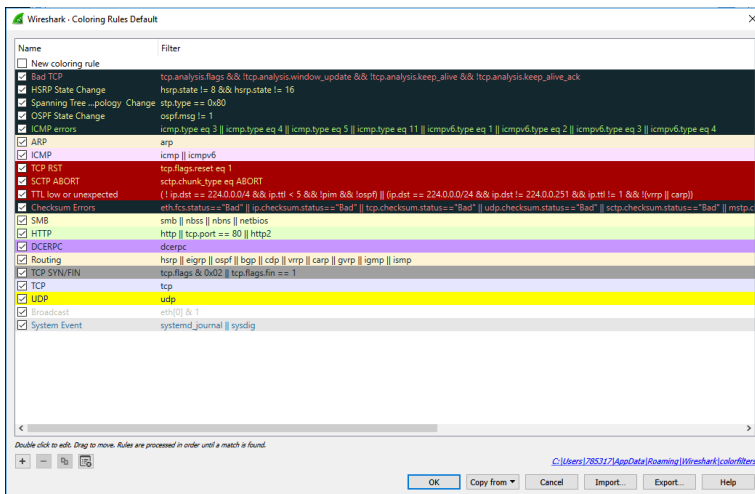
Start a new capture by clicking on the Wireshark icon at the ribbon. Please note that you can also use

Capture Menu → **Start** or **CTRL+E** keyboard shortcut to start a new capture. Select the “Continue without saving” option.

Observe TCP based packets with Blue font over Yellow background.

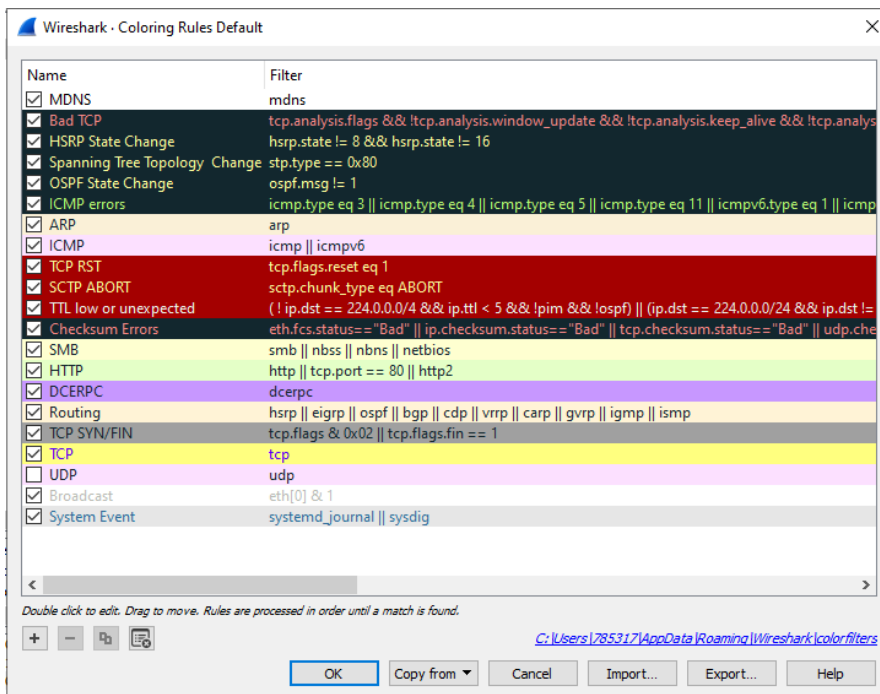


If you click on the “+” button you can add a new rule. A new empty row will be displayed at the top of the list.



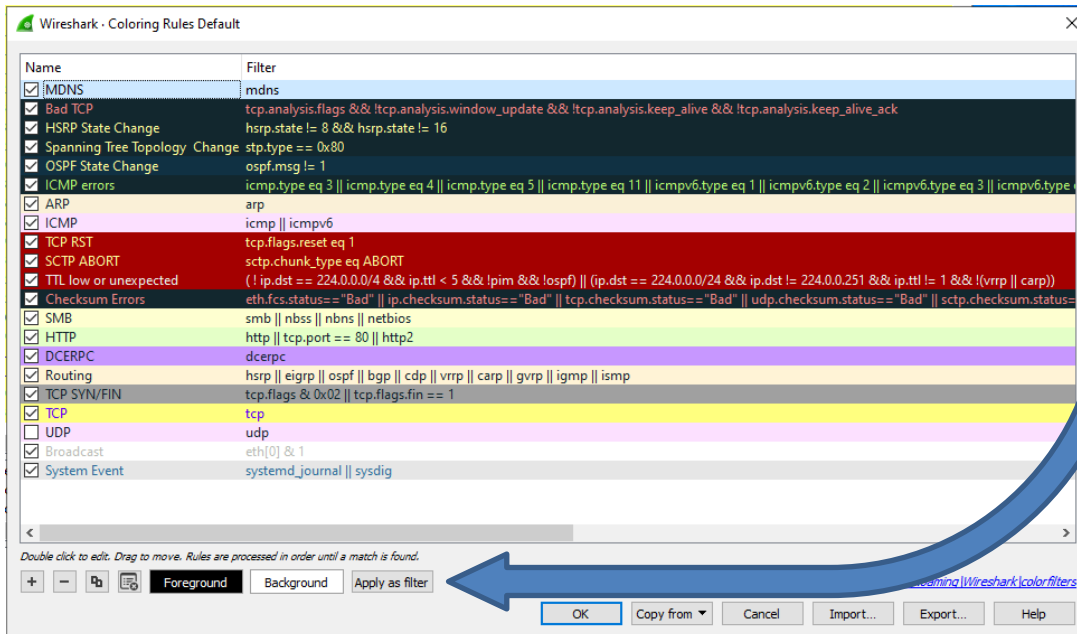
To give a Name to this rule you need to double click on the Name section (currently displays “New coloring rule”). You also need to determine a filter for this rule in the same way.

Change the Name to “MDNS” and add a filter rule by typing “mdns” at the row newly created. Tick the small box on the left hand side:



Click OK.

When you click on this rule or any existing rules you will see that different buttons at the bottom of the page are enabled. – button deletes the selected rule while  button duplicates this rule. **Foreground** and **Background** buttons change the font and background colours of the selected rule. Using Apply as filter button this rule can be copied to the packet filter line in the main screen:



Change the foreground and background of one of the rules to make it more visible. Observe this change in the Packet List Pane.

Start a new capturing and see if you can spot easily the packets belonging to this rule in real time.

HOMEWORK (if you have a computer at home):

In the Lab;

1. Start a new capture.
2. Run it for one minute and stop the capture.
3. Save it in your folder.

In the home;

1. Download and install Wireshark if it has not been installed before.
2. Start a new capture.
3. Run it for one minute and stop the capture.
4. Save it in your folder.
5. Compare it with the capture you saved in the Lab.

Discuss the following:

- **How many different protocols did you capture at the home?**
- **What are the differences?**
- **Why are there more diverse protocols in the home capture?**